INCH-POUND
MIL-PRF-3098/67E
12 January 2005
SUPERSEDING
MIL-PRF-3098/67D
27 August 1997

PERFORMANCE SPECIFICATION SHEET

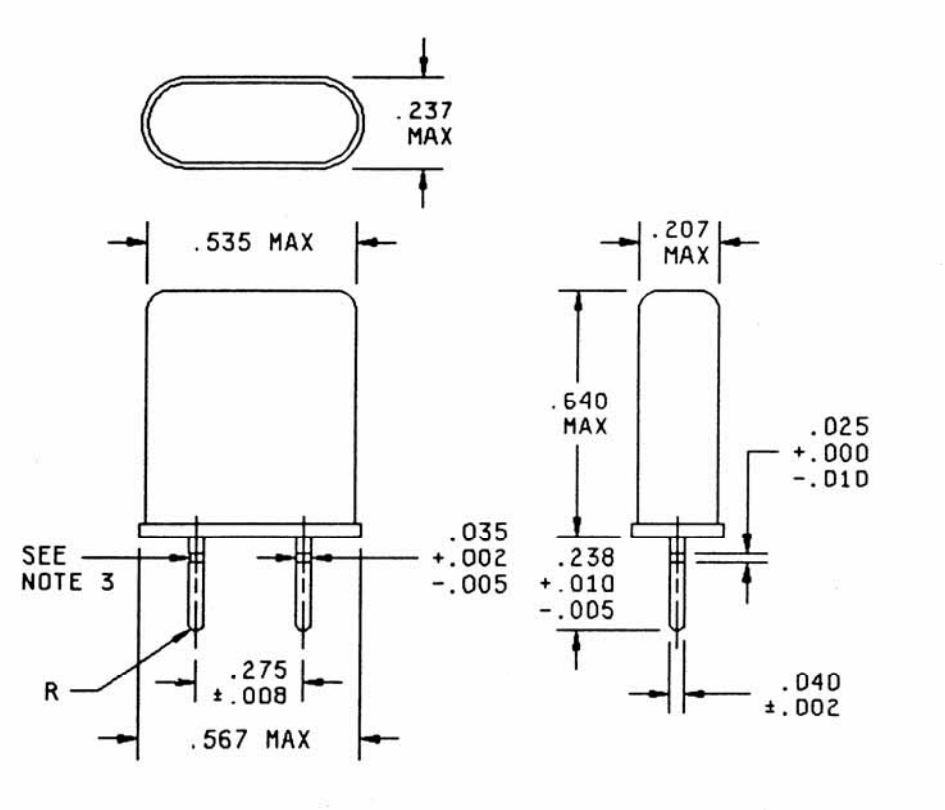
CRYSTAL UNIT, QUARTZ, CR89/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 2.12 MHz to 6.2 MHz; fundamental; noncontrolled; antiresonance.

Inches	mm
.000	0.00
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.025	0.64
.035	0.89
.040	1.02
.207	5.26
.237	6.02
.238	6.05
.275	6.99
.535	13.59
.567	14.40
.640	16.26



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin undercut may be omitted.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR89/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.12 MHz to 6.2 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See table I.

Antiresonance, load capacitance: 30.0 pF ±0.5 pF.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Frequency and equivalent resistance:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive MHz	Maximum resistance Ohms
2.12 to 2.25	500
2.25+ to 2.80	320
2.80+ to 3.41	250
3.41+ to 4.10	180
4.10+ to 5.10	120
5.10+ to 6.20	75

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5955-0797)

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